



Sadhana Devi S V 250701620-2025-CSE ▾

S2**Started on** Monday, 28 September 2026, 4:11 PM**State** Finished**Completed on** Monday, 28 September 2026, 4:12 PM**Time taken** 30 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int a[n];
9      int freq[n + 1];
10
11     for (int i = 0; i <= n; i++)
12         freq[i] = 0;
13
14     for (int i = 0; i < n; i++)
15     {
16         scanf("%d", &a[i]);
17         freq[a[i]]++;
18     }
19
20     for (int i = 1; i <= n; i++)
21     {
22         if (freq[i] > 1)
23         {
24             printf("%d", i);
25             break;
26         }
27     }
28
29     return 0;
30 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 28 September 2026, 4:12 PM**State** Finished**Completed on** Monday, 28 September 2026, 4:13 PM**Time taken** 1 min 1 sec**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

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Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main()
5  {
6      int n;
7      scanf("%d", &n);
8
9      int *a = calloc(n + 1, sizeof(int));
10
11     for (int i = 0; i < n; i++)
12     {
13         int x;
14         scanf("%d", &x);
15
16         a[x]++;
17
18         if (a[x] == 2)
19         {
20             printf("%d", x);
21             break;
22         }
23     }
24
25     free(a);
26
27     return 0;
28 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Sadhana Devi S V 250701620-2025-CSE](#) ▾**S2****Started on** Monday, 28 September 2026, 4:13 PM**State** Finished**Completed on** Monday, 28 September 2026, 4:14 PM**Time taken** 35 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int T;
6     scanf("%d", &T);
7
8     while (T--)
9     {
10         int n1, n2;
11         scanf("%d", &n1);
12
13         int a[n1];
14
15         for (int i = 0; i < n1; i++)
16             scanf("%d", &a[i]);
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for (int i = 0; i < n2; i++)
23             scanf("%d", &b[i]);
24
25         int i = 0, j = 0;
26         int first = 1;
27

```

```
28 while (i < n1 && j < n2)
29 {
30     if (a[i] == b[j])
31     {
32         if (!first)
33             printf(" ");
34
35         printf("%d", a[i]);
36         first = 0;
37         i++;
38         j++;
39     }
40     else if (a[i] < b[j])
41     {
42         i++;
43     }
44     else
45     {
46         j++;
47     }
48 }
49
50 printf("\n");
51 }
52
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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S2**Started on** Monday, 28 September 2026, 4:14 PM**State** Finished**Completed on** Monday, 28 September 2026, 4:14 PM**Time taken** 30 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int T;
6     scanf("%d", &T);
7
8     while (T--)
9     {
10         int n1, n2;
11         scanf("%d", &n1);
12
13         int a[n1];
14
15         for (int i = 0; i < n1; i++)
16             scanf("%d", &a[i]);
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for (int i = 0; i < n2; i++)

```

```

22     scanf("%d", &a[i]);
23     scanf("%d", &b[j]);
24
25     int i = 0, j = 0;
26
27     while (i < n1 && j < n2)
28     {
29         if (a[i] == b[j])
30         {
31             printf("%d ", a[i]);
32             i++;
33             j++;
34         }
35         else if (a[i] < b[j])
36         {
37             i++;
38         }
39         else
40         {
41             j++;
42         }
43     }
44
45     printf("\n");
46 }
47
48 return 0;
49 }

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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S2**Started on** Monday, 28 September 2026, 4:14 PM**State** Finished**Completed on** Monday, 28 September 2026, 4:15 PM**Time taken** 29 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, k;
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n)
18     {
19         if (i == j)
20         {
21             j++;
22         }
23         else if (a[j] - a[i] == k)
24         {
25             printf("1");
26             return 0;
27         }
28         else if (a[j] - a[i] < k)
29         {
30             j++;
31         }
32         else
33         {
34             i++;
35         }
36     }
37
38     printf("0");
39

```

```
40 |         return 0;  
41 |     }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Monday, 28 September 2026, 4:15 PM

State Finished

Completed on Monday, 28 September 2026, 4:16 PM

Time taken 28 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, k;
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n)
18     {
19         if (i == j)
20         {
21             j++;
22         }
23         else if (a[j] - a[i] == k)
24         {
25             printf("1");
26             return 0;
27         }
28         else if (a[j] - a[i] < k)
29         {
30             j++;
31         }
32         else
33         {
34             i++;
35         }
36     }
37
38     printf("0");
39

```

```
40 |         return 0;  
41 |     }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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